

Agentic-J chose BoneJ and reproduced Table 2 of the paper:
3 of 4 metrics within 0.9%, Tb.Sp +5 to +10% from an unreported setting

Task: quantify sarcomere orientation in hiPSC-derived cardiomyocytes, marked by α-actinin

Agentic-J chose TrackMate: 29 tracks, all on published cells;
mean speed 2.00 vs 2.02 µm/min published, r = 0.99

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Analysis	Time	Model output	Scientific verdict
CD177 neutrophil morphology			
Published	weeks	–	no meaningful difference
Codex CLI	13 min	0/30 features significant	no biologically meaningful conclusion due to faulty segmentation
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	no biologically meaningful conclusion due to faulty segmentation
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	no biologically meaningful conclusion due to faulty segmentation
Conventional Agentic-J + human	488 min	0/11 tested significant 34 features extracted	no meaningful difference detected, needs validation

Analysis	Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei	Task 2 proliferation in TUNEL+ hosts
Leishmania infection: TUNEL and proliferation			
Published	not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL− parasites
Codex CLI	18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 1.68; 10 of 15 parasites
Exploratory Agentic-J	22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
Conventional Agentic-J	55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
Conventional Agentic-J + human	35 / 45 min	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 2.36 (published data 2.36); 15/15

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)











